



ΚΥΠΡΙΑΚΗ ΔΗΜΟΚΡΑΤΙΑ
ΥΠΟΥΡΓΕΙΟ ΜΕΤΑΦΟΡΩΝ,
ΕΠΙΚΟΙΝΩΝΙΩΝ ΚΑΙ ΕΡΓΩΝ

ΤΜΗΜΑ ΟΔΙΚΩΝ
ΜΕΤΑΦΟΡΩΝ
ΛΕΥΚΩΣΙΑ 1425 - ΚΥΠΡΟΣ

COMMUNICATION



concerning: ⁽¹⁾ APPROVAL GRANTED
~~APPROVAL EXTENDED~~
~~APPROVAL REFUSED~~
~~APPROVAL WITHDRAWN~~
~~PRODUCTION DEFINITELY DISCONTINUED~~

with regard to gaseous and particulate pollutant emission pursuant to Regulation (EU) 2016/1628, as last amended by (Commission Delegated) ⁽¹⁾ Regulation 2022/992/EU ⁽¹⁾ ⁽²⁾ ~~(of the European Parliament and of the Council)~~ ⁽¹⁾

EU type approval No⁽³⁾: **e49*2016/1628*2022/992SRA1/P*1518*00**

Reason for ~~extension/refusal/withdrawal~~ ⁽¹⁾

SECTION I

- | | | |
|-------|--|---|
| 1.1. | Make (trade name(s) of manufacturer) | : SAIBOYUHU |
| 1.2. | Commercial name(s) (if applicable) | : not applicable |
| 1.3. | Company name and address of manufacturer | : Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village, Qingmuguan Town, Shapingba District, Chongqing, China |
| 1.4. | Name and address of manufacturer's authorised representative (if any) | : Liu Lu
MALGRATE (LC) VIA ROMA 62 CAP 23864 |
| 1.5. | Name(s) and address(es) of assembly/manufacture plant(s) | : Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village, Qingmuguan Town, Shapingba District, Chongqing, China |
| 1.6. | Engine type designation/ engine family designation ⁽¹⁾ | : ZT-170F/P |
| 1.7. | Category and sub-category of the engine type/ engine family ⁽¹⁾ ⁽⁴⁾ | : Category: NRS
Sub-category: NRS-vr-1a |
| 1.8. | Emissions durability period category | : not applicable /Cat 1/Cat 2/Cat 3 ⁽⁴⁾ |
| 1.9. | Emissions stage | : V/ SPE ⁽¹⁾ |
| 1.10. | Engine for snow throwers ⁽⁵⁾ | : yes /no ⁽¹⁾ |



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SECTION II

1. Technical service responsible for carrying out the tests : CETOC Technical Service s.r.l.
Via della Bufalotta, 374, 00139 Roma
2. Date(s) of the test report(s) : 06/08/2025
3. Number of test report(s) : CN-132-25-304-NRM25-35842-IR

SECTION III

The undersigned hereby certifies the accuracy of the manufacturer's description in the attached information document of the engine type/~~engine family~~⁽¹⁾ described above, for which one or more representative samples, selected by the approval authority, have been submitted as prototypes and that the attached test results apply to the engine type/~~engine family~~⁽¹⁾.

1. The engine type/~~engine family~~⁽¹⁾ meets/~~does not meet~~⁽¹⁾ the requirements laid down in Regulation (EU) 2016/1628.
 2. The approval is granted/~~extended/refused/withdrawn~~⁽¹⁾
 3. The approval is granted in accordance with Article 35 of Regulation (EU) 2016/1628 and the validity of the approval is thus limited to dd/mm/yyyy⁽³⁾: not applicable
 4. Restrictions to validity^{(3) (6)} : not applicable
 5. Exemptions applied^{(3) (6)} : not applicable
- Place : Nicosia, Cyprus
- Date : 12/11/2025
- Name and signature :

Georgios A. Louca
Road Transport Officer

Attachments:

- Information package
- Test report(s)



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Where applicable, the name(s) and specimen(s) of the signature(s) of the person(s)
authorised to sign statement of conformity and a statement of their position in the company

Where applicable, a completed specimen of a statement of conformity





ADDENDUM

to EU type-approval number: e49*2016/1628*2022/992SRA1/P*1518*00

PART A – CHARACTERISTICS OF THE ENGINE TYPE/ENGINE FAMILY ⁽¹⁾

2. Common design parameters of the engine type/engine family ⁽¹⁾

- 2.1. Combustion cycle : four stroke cycle/~~two stroke cycle/rotary/~~
~~other (describe)~~ ⁽¹⁾
- 2.2. Ignition Type : ~~Compression ignition~~/spark ignition ⁽¹⁾
- 2.3.1. Position of the cylinders in the block : ~~V/in line/radial~~/other (describe) ⁽¹⁾:single
- 2.6. Main Cooling medium : Air/~~Water~~/~~Oil~~ ⁽⁴⁾
- 2.7. Method of air aspiration : naturally aspirated/~~pressure charged/pressure charged~~
~~with charge cooler~~ ⁽⁴⁾
- 2.8.1. Fuel type(s) : ~~Diesel (non road gas oil)/Ethanol for dedicated~~
~~compression ignition engines (ED95)/~~
Petrol (E10)/Ethanol (E85)/(Natural gas/
Biomethane)/Liquid Petroleum Gas (LPG) ⁽⁴⁾
- 2.8.1.1. Sub fuel type : ~~Universal fuel — high calorific fuel (H gas) and low~~
~~calorific fuel (L gas)/Restricted fuel —~~
(Natural gas/Biomethane only) ~~high calorific fuel (H gas)/Restricted fuel —~~
~~low calorific fuel (L gas)/Fuel specific (LNG)~~ ⁽¹⁾
- 2.8.2. Fuelling arrangement : Liquid-fuel only/~~Gaseous fuel only/Dual fuel type~~
~~1A/Dual fuel type 1B/Dual fuel type 2A/~~
~~Dual fuel type 2B/Dual fuel type 3B~~ ⁽¹⁾
- 2.8.3. List of additional fuels compatible with : not applicable
use by the engine declared by the
manufacturer in accordance with point 1
of Annex I to Delegated Regulation (EU)
2017/654
(provide reference to recognised standard
or specification)
- 2.8.4. Lubricant added to fuel : ~~Yes~~/No ⁽¹⁾
- 2.8.5. Fuel supply type : ~~Pump (high pressure) line and injector/in line pump or~~
~~distributor pump/Unit injector/Common~~
~~rail/Carburettor/port injector/direct injector/~~
~~Mixing unit/other (specify)~~ ⁽⁴⁾
- 2.9. Engine management systems : mechanical/~~electronic control strategy~~ ⁽⁴⁾



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- 2.10. *Miscellaneous devices* : ~~Yes~~/No ⁽¹⁾
- 2.10.1. Exhaust gas recirculation (EGR) : ~~Yes~~/No ⁽¹⁾
- 2.10.2. Water injection : ~~Yes~~/No ⁽¹⁾
- 2.10.3. Air injection : ~~Yes~~/No ⁽¹⁾
- 2.10.4. Others (specify) : not applicable
- 2.11. *Exhaust after-treatment system* : ~~Yes~~/No ⁽¹⁾
- 2.11.1. Oxidation catalyst : ~~Yes~~/No ⁽¹⁾
- 2.11.2. DeNO_x system with selective reduction of NO_x : ~~Yes~~/No ⁽¹⁾
- (addition of reducing agent)
- 2.11.3. Other DeNO_x systems : ~~Yes~~/No ⁽¹⁾
- 2.11.4. Three-way catalyst combining oxidation and NO_x reduction : ~~Yes~~/No ⁽¹⁾
- 2.11.5. Particulate after-treatment system with passive regeneration : ~~Yes~~/No ⁽¹⁾
- 2.11.6. Particulate after-treatment system with active regeneration : ~~Yes~~/No ⁽¹⁾
- 2.11.7. Other particulate after-treatment systems : ~~Yes~~/No ⁽¹⁾
- 2.11.8. Other after-treatment devices (specify) : not applicable
- 2.11.9. Other devices or features that have a strong influence on emissions (specify) : not applicable

3. Essential characteristics of the engine type(s)

Item No	Item description	Parent engine/ Engine type	Engine types within the family (if applicable)
3.1.1.	Engine Type Designation:	ZT-170F/P	not applicable
3.1.2.	Engine type designation shown on engine mark:	Yes	not applicable
3.1.3.	Location of the manufacturer's statutory marking:	Refer to drawing No.001	not applicable
3.2.1.	Declared rated speed (rpm):	3600	not applicable
3.2.1.2.	Declared rated net power (kW):	3.6	not applicable
3.2.2.	Maximum power speed (rpm):	3600	not applicable



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3.2.2.2.	Maximum net power (kW):	3.6	not applicable
3.2.3.	Declare maximum torque speed (rpm):	2800	not applicable
3.2.3.2.	Declared maximum torque (Nm):	12.0	not applicable
3.6.3.	Number of cylinders:	1	not applicable
3.6.4.	Engine total swept volume (cm ³):	212	not applicable
3.8.5.	Device for recycling crankcase gases: Yes/No ⁽¹⁾ :	No	not applicable
3.11.3.12.	Consumable reagent: Yes/No ⁽¹⁾ :	No	not applicable
3.11.3.12.1.	Type and concentration of reagent needed for catalytic action:	not applicable	not applicable
3.11.3.13.	NO _x sensor(s): Yes/No ⁽¹⁾ :	No	not applicable
3.11.3.14.	Oxygen sensor: Yes/No ⁽¹⁾ :	No	not applicable
3.11.4.7.	Fuel borne catalyst (FBC): Yes/No ⁽¹⁾ :	No	not applicable
Particular conditions to be respected in the installation of the engine on non-road mobile machinery:			
3.8.1.1.	Maximum allowable intake depression at 100 % engine speed and at 100 % load (kPa) with clean air cleaner:	2.0	not applicable
3.8.3.2.	Maximum charge air cooler outlet temperature at 100 % speed and 100 % load (deg. C):	not applicable	not applicable
3.8.3.3.	Maximum allowable pressure drop across charge cooler at 100 % engine speed and at 100 % load (kPa) (if applicable):	not applicable	not applicable
3.9.3.	Maximum permissible exhaust gas back-pressure at 100 % engine speed and at 100 % load (kPa):	5.5	not applicable
3.9.3.1.	Location of measurement:	The first chamber of muffler	not applicable
3.11.1.2.	Maximum temperature drop from exhaust system or turbine outlet to first exhaust after-treatment system (deg. C) if stated:	not applicable	not applicable
3.11.1.2.1.	Test conditions for measurement:	not applicable	not applicable

PART B – TEST RESULTS

3.8. Manufacturer intends to use ECU torque signal for in-service monitoring : ~~Yes~~/No ⁽¹⁾



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3.8.1. Dynamometer torque greater than or equal to 0,93 x ECU torque : ~~Yes~~/No ⁽¹⁾

3.8.2. ECU torque correction factor in case that dynamometer torque less than 0,93 x ECU torque : not applicable

11.1. Cycle emission results:

Emissions	CO (g/kWh)	HC (g/kWh)	NO _x (g/kWh)	HC+NO _x (g/kWh)	PM (g/kWh)	PN #/kWh	Test Cycle ⁽²⁾
NRSC final result with DF	351.18	7.05	2.13	9.18	--	--	G2
NRTC final test result with DF	--	--	--	--	--	--	--

11.2. CO₂ result : 968.21 g/kWh

11.3. In service monitoring reference values ⁽³⁾ : not applicable

11.3.1. Reference work (kWh) : not applicable

11.3.2. Reference CO₂ mass (g) : not applicable



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INDEX TO TYPE-APPROVAL

EU type-approval number: e49*2016/1628*2022/992SRA1/P*1518*00

Manufacturer's name: Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.

Type: ZT-170F/P

Report: CN-132-25-304-NRM25-35842-IR

Information document: DOC. No. ZT-170F/P



NRMM - Engines (Stage V)

0. Legislation:

- 0.1. Requirements according to : Regulation (EU) 2016/1628, amended by Regulation (EU) 2022/992
Delegated Regulation (EU) 2017/654, as last amended by
Regulation (EU) 2021/1398
Delegated Regulation (EU) 2017/655, as last amended by
Regulation (EU) 2022/2387
Implementing Regulation (EU) 2017/656, as last amended by
Regulation (EU) 2018/988

1. General

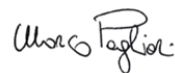
- 1.1. Reason for Inspection Report : New approval / ~~Extension of approval~~ / ~~Test report only~~ / ~~COP~~
1.1.1. Previous Type Approval Number : N/A
1.2. Manufacturer's Representative(s) : Li Wei
1.3. CETOC TS Representative(s) : Anne Zhu
1.4. Location of test : Nanjing Depurate Environmental Technology Co., Ltd.
No.18 Youshan Road, Gaochun Economic Development Zone,
Nanjing, China
1.5. Date of test : 28/07/2025-05/08/2025

2. Manufacturer Details

- 2.1. Make : SAIBOYUHU
2.2. Type : ZT-170F/P
2.3. Variant/Version : N/A
2.4. Commercial Name : N/A
2.5. Category : NRS-vr-1a
2.6. Name and Address of manufacturer : Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village,
Qingmuguan Town, Shapingba District, Chongqing, China

3. Conclusion:

- 3.1. Final conclusion of the inspection : The above mentioned type was tested in accordance with the above mentioned legislation and was found to comply in all respects. This Inspection report relates only to the items tested.

Signature :		
Name :	Anne Zhu	Marco Pagliari
Position :	Type Approval Engineer	Tech. Mgr.
Place and date :	Hangzhou, 06/08/2025	Roma, 06/08/2025

4. List of Appendixes:

Appendix Nr.	Page Nr.	Subject
Appendix 1	Page 2	: Test report history
Appendix 2	Page 2	: General specification
Appendix 3	Page 3	: Inspection results
Appendix 4	Page 13	: Test Photos

APPENDIX 1 - TEST REPORT HISTORY

List this report and previous reports, with extension details.

Inspection Report Number	Reason for Extension	Date of Issue
CN-132-25-304-NRM25-35842-IR	New approval	06/08/2025

APPENDIX 2 – GENERAL SPECIFICATION

1. **Worst Case Rationale** : Single specification

2. **Significant Interpretations, Alternative Test Methods, New Technologies** : N/A

3. **Engine / Parts Specification**

Engine type	ZT-170F/P
Engine Serial Number	250916435
Carburetor	mucan / 170F
Spark plug	DNNID / F7TC
After-treatment Catalyst	N/A

4. **Facility and Equipment Checks**

4.1. Calibration certificates checked and valid, recorded in the following table : Conform

4.2. All instruments are equipped with identification label : Yes

4.3. Calibration certificates are complete of calibration-chain with detailed information regarding primary used. : Yes

Equipment	Serial / Certificate No.	Calibration due*
Dynamometer	T2506250270069	30/06/2026
Fuel flow meter	T2506250270059	30/06/2026
Gas analyser	T2507070260012	09/07/2026
Weather station	T2506250270086	06/07/2026
Barometer	H2025-0068137	06/07/2026

*Specify calibrated date + (interval) or calibration due date.

APPENDIX 3 – INSPECTION RESULTS

1. General Information

1.1.	Make(s) (trade name(s) of manufacturer):	SAIBOYUHU
1.2.	Commercial names(s) (if applicable):	N/A
1.3.	Company name and address of manufacturer:	Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd. The second and third floors of No. 16, Yijian 2nd Village, Qingmuguan Town, Shapingba District, Chongqing, China
1.4.	Name of technical service:	CETOC Technical Service s.r.l.
1.5.	Address of technical service:	Via della Bufalotta 374 - 00139 ROMA, ITALY
1.6.	Location of test:	Nanjing Depurate Environmental Technology Co., Ltd. No.18 Youshan Road, Gaochun Economic Development Zone, Nanjing, China
1.7.	Date of test:	28/07/2025-05/08/2025
1.8.	Test report number:	CN-132-25-304-NRM25-35842-IR
1.9.	Information document reference number (if available):	DOC. No. ZT-170F/P
1.10.	Test report type:	Primary test* *Delete as appropriate.
1.10.1	Description of the purpose of the test:	New approval test

2. General engine information (test engine)

2.1.	Engine type designation/engine family designation/ET:	ZT-170F/P
2.2.	Engine identification number:	250916435
2.3.	Engine category and subcategory:	NRS-vr-1a *Delete as appropriate.

3. Documentation and information check list (primary test only)

3.1.	Engine mapping documentation reference:	ZT-170F/P Engine map
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3.2.	Deterioration factor determination documentation reference:	ZT-170F/P Durability record
3.3.	Infrequent regeneration factors determination documentation reference, where applicable:	N/A
3.4.	NOx control diagnostic determination documentation reference, where applicable:	N/A
3.5.	Particulate control diagnostic determination documentation reference, where applicable:	N/A
3.6.	For engine types and engine families that use an ECU as part of the emission control system anti-tampering declaration documentation reference:	N/A
3.7.	For engine types and engine families that use mechanical devices as part of the emission control system anti-tampering and adjustable parameters declaration and demonstration documentation reference:	Manufacturer's statement on No Use of "Defeat Device"
3.8.	Manufacturer intends to use ECU torque signal for in-service monitoring:	No* <i>*Delete as appropriate.</i>
3.8.1.	Dynamometer torque greater than or equal to 0.93 x ECU torque:	No* <i>*Delete as appropriate.</i>
3.8.2.	ECU torque correction factor in case that dynamometer torque less than 0.93 x ECU torque:	N/A
4.	Reference fuel(s) used for test (complete relevant subparagraphs(s))	
4.1.	<i>Liquid fuel for spark-ignition engines</i>	
4.1.1.	Make:	SINOPEC
4.1.2.	Type:	Petrol(E10)
4.1.3.	Octane number RON:	97.2
4.1.4.	Octane number MON:	86.2
4.1.5.	Ethanol content (%):	9.72
4.1.6.	Density at 15 Deg.C (kg/m ³)	751.6

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4.2. *Liquid fuel for compression-ignition engines*

4.2.1.	Make:	N/A
4.2.2.	Type:	N/A
4.2.3.	Cetane number:	N/A
4.2.4.	Fame content (%):	N/A
4.2.5.	Density at 15 Deg.C (kg/m ³)	N/A

4.3 *Gaseous fuel – LPG*

4.3.1.	Make:	N/A
4.3.2.	Type:	N/A
4.3.3.	Reference fuel type:	N/A <i>*Delete as appropriate.</i>
4.3.4.	Octane number MON:	N/A

4.4. *Gaseous fuel – Methane/biomethane*

4.4.1.	Reference fuel type:	N/A <i>*Delete as appropriate.</i>
4.4.2.	Source of reference gas:	N/A <i>*Delete as appropriate.</i>
4.4.3.	For specific reference fuel	
4.4.3.1.	Make:	N/A
4.4.3.2.	Type:	N/A
4.4.4.	For pipeline gas with admixture	
4.4.4.1.	Admixture(s):	N/A <i>*Delete as appropriate.</i>
4.4.4.2.	The value of S _λ for the resulting fuel blend:	N/A
4.4.4.3.	The Methane Number (MN) of the resulting fuel blend:	N/A

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4.5. Dual fuel engine (in addition to relevant sections above)

4.5.1. Gas energy ratio on test cycle: N/A

5. Lubricant

5.1. Make(s): MOBIL

5.2. Type(s): FC

5.3. SAE viscosity: 5W/30

5.4. Lubricant and fuel are mixed: No

**Delete as appropriate.*

5.4.1. Percentage of oil in mixture: N/A

6. Engine Speed

6.1. 100 % speed (rpm): 3600

6.1.1. 100 % speed determined by: Declared rated speed

**Delete as appropriate.*

6.1.2. Adjusted MTS if applicable (rpm): N/A

6.2. Intermediate speed: N/A

6.2.1. Intermediate speed determined by: N/A

**Delete as appropriate.*

6.3. Idle speed: 1600

7. Engine Power

7.1. Engine driven equipment (if applicable)

~~7.1.1. Power absorbed at indicated engine speeds by necessary auxiliaries for engine operation that cannot be fitted for the test (as specified by the manufacturer) to be stated in Table 1:~~

~~Table 1
Power absorbed by engine auxiliaries~~

Auxiliary type and identifying details	Power absorbed by auxiliary (kW) at indicated engine speed (complete relevant columns)						
	Idle	63%	80%	91%	Inter-mediate	Max. power	100%
	-	-	-	-	-	-	-
	-	-	-	-	-	-	-
	-	-	-	-	-	-	-
Total ($P_{t,t}$)	-	-	-	-	-	-	-

~~7.1.2.~~ ~~Power absorbed at indicated engine speeds by auxiliaries linked with the operation of the non road mobile machinery that cannot be removed for the test (as specified by the manufacturer) to be stated in Table 2:~~

Table 2
Power absorbed by non-road mobile machinery auxiliaries

Auxiliary type and identifying details	Power absorbed by auxiliary (kW) at indicated engine speed (complete relevant columns)						
	Idle	63%	80%	91%	Inter-mediate	Max. power	100%
	-	-	-	-	-	-	-
	-	-	-	-	-	-	-
	-	-	-	-	-	-	-
Total ($P_{r,i}$)	-	-	-	-	-	-	-

7.2. Engine net power to be stated in Table 3:

Table 3
Engine net power

Condition	Engine net power (kW) at indicated engine speed (complete relevant columns)		
	Intermediate	Max. power	100%
Reference power measured at specified test speed ($P_{m,i}$)	-	3.6	3.6
Total auxiliary power from Table 1 ($P_{f,i}$)	-	-	-
Total auxiliary power from Table 2 ($P_{r,i}$)	-	-	-
Engine net power $P_i = P_{m,i} - P_{f,i} + P_{r,i}$	-	3.6	3.6

8. **Conditions at test**

- 8.1. f_a within range 0.93 to 1.07: Yes
*Delete as appropriate.
- 8.1.1. If f_a is not within specified range state altitude of test facility and dry atmospheric pressure: N/A
- 8.2. Applicable intake air temperature range: 20 to 30
*Delete as appropriate.

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9. **Information concerning the conduct of the NRSC test:**

9.1. Cycle (mark cycle used with X) to be stated in Table 4:

Table 4
NRSC test cycle

Cycle	C1	C2	D2	E2	E3	F	G1	G2	G3	H
Discrete mode	-	-	-	-	-	-	-	X	-	-
RMC	-	-	-	-	-	-	-	-	-	-

9.2. Dynamometer setting (kW) to be stated in Table 5:

Table 5
Dynamometer setting

% Load at point or % of rated power (as applicable)	Dynamometer setting (kW) at indicated engine speed after adjustment for auxiliary power ⁽¹⁾ (complete relevant columns)					
	Idle	63%	80%	91%	Intermediate	100%
0%	0.04	-	-	-	-	-
5%	-	-	-	-	-	-
10%	-	-	-	-	-	0.36
25%	-	-	-	-	-	0.90
50%	-	-	-	-	-	1.80
75%	-	-	-	-	-	2.70
100%	-	-	-	-	-	3.60

⁽¹⁾ The dynamometer setting shall be determined using the procedure set out in point 7.7.1.3 of Annex VI to Delegated Regulation (EU) 2017/654. The auxiliary power in that point shall be determined using the total values set out in Tables 1 and 2 of this Appendix.

9.3. NRSC Emissions results

9.3.1. Deterioration Factor (DF):

Calculated / ~~Assigned~~*

*Strike-through, as appropriate.

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9.3.2. DF values and the cycle weighted emissions results to be stated in Table 6:

Note: In the event that a discrete mode NRSC is run where the K_{ru} or K_{rd} factors have been established for individual modes then a table showing each mode and the applied K_{ru} or K_{rd} should replace the shown table.

Table 6

NRSC cycle DF values and weighted emissions results

Test fuel: E10

DF	CO	HC	NOx	HC+NOx	PM	PN
mult/add	1.021	1.135	1.000	1.086	-	-
Emissions	CO (g/kWh)	HC (g/kWh)	NOx (g/kWh)	HC+NOx (g/kWh)	PM (g/kWh)	PN (#/kWh)
Test result with/without regeneration	343.98	6.21	2.24	8.45	-	-
k_{ru}/k_{rd} mult/add	-	-	-	-	-	-
test result with infrequent regeneration adjustment factors (IRAFs)	-	-	-	-	-	-
Final test result with DF	351.18	7.05	2.13	9.18	-	-

9.3.3. Cycle weighted CO₂ (g/kWh): 968.21

9.3.4. Cycle weighted NH₃ (ppm): N/A

9.4. Additional control area test points (if applicable) to be stated in Table 7:

Table 7

Additional control area test points

Emissions at test point	Engine speed	Load (%)	CO (g/kWh)	HC (g/kWh)	NOx (g/kWh)	HC+NOx (g/kWh)	PM (g/kWh)	PN (n/kWh)
Test result 1	-	-	-	-	-	-	-	-
Test result 2	-	-	-	-	-	-	-	-
Test result 3	-	-	-	-	-	-	-	-

9.5. Sampling systems used for the NRSC test:

9.5.1. Gaseous emissions: N/A

9.5.2. PM: N/A

9.5.2.1. Method: Single / multiple filter*

*Strike-through, as appropriate.

9.5.3. Particle number: N/A

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10. **Information concerning the conduct of the transient test (if applicable)**

10.1. Cycle (mark cycle with X) to be stated in Table 8:

Table 8

Transient test cycle

NRTC	-
LSI-NRTC	-

10.2. Transient test deterioration factors:

10.2.1. Deterioration Factor (DF):

Calculated / Fixed*

*Strike through, as appropriate.

10.2.2. DF values and the emissions results to be stated in Table 9 or Table 10

10.3. NRTC emissions results:

Table 9

DF values and the emission results for NRTC

DF mult/add	CO	HC	NOx	HC+NOx	PM	PN
	-	-	-	-	-	-
Emissions	CO (g/kWh)	HC (g/kWh)	NOx (g/kWh)	HC+NOx (g/kWh)	PM (g/kWh)	PN (#/kWh)
Cold start	-	-	-	-	-	-
Hot start test result with/ without regeneration	-	-	-	-	-	-
Weighted test result	-	-	-	-	-	-
k_{tu}/k_{td} mult/add	-	-	-	-	-	-
Weighted test result with IRAFs	-	-	-	-	-	-
Final test result with DF	-	-	-	-	-	-

10.3.1. Hot cycle CO₂ (g/kWh):

-

10.3.2. Cycle weighted NH₃ (ppm):

-

10.3.3. Cycle work for hot start test (kWh):

-

10.3.4. Cycle CO₂ for hot start test (g):

-

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10.4. LSI NRTC emission results

Table 10

DF values and the emission results for NRTC-LSI

Test Fuel: D

DF mult/add	CO	HC	NOx	HC+NOx	PM	PN
	-	-	-	-	-	-
Emissions	CO (g/kWh)	HC (g/kWh)	NOx (g/kWh)	HC+NOx (g/kWh)	PM (g/kWh)	PN (#/kWh)
Test result with/without regeneration	-	-	-	-	-	-
k_{tr}/k_{td} mult/add	-	-	-	-	-	-
Test result with IRAFs	-	-	-	-	-	-
Final test result with DF	-	-	-	-	-	-

10.4.1.	Cycle CO ₂ (g/kWh):	-
10.4.2.	Cycle average NH ₃ (ppm):	-
10.4.3.	Cycle work (kWh):	-
10.4.4.	Cycle CO ₂ (g):	-
10.5.	Sampling system used for the NRTC test:	
10.5.1.	Gaseous emissions:	-
10.5.2.	PM:	-
10.5.3.	Particle number:	-

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11. Final emissions results

11.1. Cycle emissions results to be stated in Table 11.

Table 11
Final emissions results

Emissions	CO (g/kWh)	HC (g/kWh)	NOx (g/kWh)	HC+NOx (g/kWh)	PM (g/kWh)	PN (#/kWh)	Test Cycle ⁽¹⁾
NRSC final result with DF	351.18	7.05	2.13	9.18	-	-	G2
Final transient test result with DF	-	-	-	-	-	-	-

11.2. CO₂ result: 968.21 g/kwh

11.3. In service monitoring reference values

11.3.1. Reference work (kWh): N/A

11.3.2. Reference CO₂ mass (g): N/A

Remarks

None

Note: CETOC TS apply measurement uncertainty to calibrated items but not test results.

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APPENDIX 4 – PHOTOS



Content of Information Document

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INFORMATION DOCUMENT

Relating to type-approval and referring to measures against the emission of gaseous and particulate pollutants from internal combustion engine to be installed in non –road mobile machinery

(According to Regulation (EU) 2016/1628 as last amend by Regulation (EU) 2022/992)

Part A

1. GENERAL INFORMATION

- | | | | |
|------|--|---|--|
| 1.1 | Make (trade name(s) of manufacturer) | : | SAIBOYUHU |
| 1.2 | Commercial name(s) (if applicable) | : | Not applicable |
| 1.3 | Company name and address of manufacturer | : | Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village,
Qingmuguan Town, Shapingba District, Chongqing, China |
| 1.4 | Name and address of manufacturer's authorised representative (if any) | : | Liu Lu
MALGRATE (LC) VIA ROMA 62 CAP 23864 |
| 1.5 | Name(s) and address(es) of assembly/manufacture plant(s) | : | Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village,
Qingmuguan Town, Shapingba District, Chongqing, China |
| 1.6 | Engine type designation/ engine family -
designation/FT | : | ZT-170F/P |
| 1.7 | Category and sub-category of the engine
type/ engine family | : | Category : NRS
Sub-category: NRS-vr-1a |
| 1.8 | Emissions durability period category | : | Cat 1 |
| 1.9 | Emissions stage | : | V |
| 1.10 | In case of NRS <19 kW only, engine family
consisting exclusively of engine types for
snow throwers | : | Not applicable |
| 1.11 | Reference power is | : | Maximum net power |
| 1.12 | Primary NRSC test cycle | : | G2 |

- | | | | |
|--------|---|---|----------------|
| 1.12.1 | In case of variable speed IWP category only, Additional propulsion test cycle | : | Not applicable |
| 1.12.2 | In case of IWP category only, additional auxiliary NRSC test cycle | : | Not applicable |
| 1.13 | Transient test cycle | : | Not applicable |
| 1.14 | Restrictions on use (if applicable) | : | Not applicable |

Part B

2. COMMON DESIGN PARAMETERS OF ENGINE FAMILY⁽¹⁾

- | | | | |
|--|---|---|-------------------------|
| 2.1 | Combustion Cycle | : | Four stroke cycle |
| 2.2 | Ignition Type | : | Spark ignition |
| 2.3 Configuration of the cylinders | | | |
| 2.3.1 | Position of the cylinders in the block | : | Single |
| 2.3.2 | Bore centre to centre dimension (mm) | : | Not applicable |
| 2.4. Combustion chamber type/design | | | |
| 2.4.1 | Open chamber/divided chamber/other(specify) | : | Open chamber |
| 2.4.2 | Valve and porting configuration | : | Refer to drawing No.002 |
| 2.4.3 | Number of valves per cylinder | : | 2 |
| 2.5 | Range of swept volume per cylinder (cm ³) | : | See 3.6.4. Part C |
| 2.6 | Main Cooling medium | : | Air |
| 2.7 | Method of air aspiration | : | Naturally aspirated |
| 2.8. Fuel | | | |
| 2.8.1 | Fuel Type | : | Petrol (E10) |
| 2.8.1.1 | Sub Fuel type (Natural gas/Biomethane only) | : | Not applicable |
| 2.8.2 | Fuelling arrangement | : | Liquid-fuel only |

2.8.3	list of additional fuels, fuel mixtures or emulsions suitable for use by the engine, as declared by the manufacturer in accordance with point 1.2.3 of Annex I to Delegated Regulation (EU) 2022/992 (provide reference to recognised standard or specification)	:	Not applicable
2.8.4	Lubricant added to fuel	:	Not applicable
2.8.4.1	Specification	:	Not applicable
2.8.4.2	Ratio of fuel to oil	:	Not applicable
2.8.5	Fuel supply type: Pump (high pressure) line and injector/in-line pump or distributor pump/Unit injector/Common rail/Carburetor/port injector/direct injector/Mixing unit/other(specify)	:	Carburetor
2.9	Engine management systems	:	Mechanical
2.10	Miscellaneous devices		
2.10.1	Exhaust gas recirculation	:	No (if yes, complete section 3.10.1. and provide a schematic diagram of the location and order of the devices)
2.10.2	Water injection	:	No (if yes, complete section 3.10.2. and provide a schematic diagram of the location and order of the devices)
2.10.3	Air injection	:	No (if yes, complete section 3.10.3. and provide a schematic diagram of the location and order of the devices)
2.10.4	Others	:	No (if yes, complete section 3.10.4. and provide a schematic diagram of the location and order of the devices)
2.11	Exhaust after-treatment system	:	No (if yes provide a schematic diagram of the location and order of the devices)
2.11.1	Oxidation catalyst	:	No (if yes, complete section 3.11.2.)
2.11.2	DeNOx system with selective reduction of NOx (addition of reducing agent)	:	No (if yes, complete section 3.11.3.)

2.11.3	Other DeNOx systems	:	No (if yes, complete section 3.11.3.)
2.11.4	Three-way catalyst combining oxidation and NOx reduction	:	No (if yes, complete section 3.11.3.)
2.11.5	Particulate after-treatment system with passive regeneration	:	No (if yes, complete section 3.11.4.)
2.11.5.1.	Wall flow/non-wall-flow		Not applicable
2.11.6	Particulate after-treatment system with active regeneration	:	No (if yes, complete section 3.11.4.)
2.11.6.1.	Wall flow/non-wall-flow		Not applicable
2.11.7	Other particulate after-treatment systems	:	No (if yes, complete section 3.11.4.)
2.11.8	Other after-treatment devices (specify)	:	No (if yes, complete section 3.11.5.)
2.11.9	Other devices or features that have a strong influence on emissions	:	Not applicable

Part C

3 ESSENTIAL CHARACTERISTICS OF THE ENGINE TYPE(S)

Item Number	Item Description	Test	Installation	Homologation	Parent engine/ Engine type	Engine types within the engine family (if applicable)	Explanatory notes (not included in document)
						Type 2	
3.1	Engine Identification						
3.1.1	Engine type designation			X	ZT-170F/P		
3.1.2	Engine type designation shown on engine marking: yes/no			X	Yes		
3.1.3	Location of the statutory marking			X	Refer to drawing No.001		
3.1.4	Method of attachment of the statutory marking			X	Engrave and/or Paste		
3.1.5	Drawings of the location of the engine identification number (complete example with dimensions)			X	Refer to drawing No.001		
3.2	Performance Parameters						
3.2.1	Declared rated speed (rpm)	X			3600		
3.2.1.1	Fuel delivery/stroke (mm ³) for diesel engine, fuel flow (g/h) for other engines, at rated net power			X	1150		
3.2.1.2	Declared rated net power (kW)	X			3.6		
3.2.2	Maximum power speed(rpm)			X	3600		If different from rated speed
3.2.2.1	Fuel delivery/stroke (mm ³) for diesel engine, fuel flow (g/h) for other engines, at maximum net power			X	1150		
3.2.2.2	Maximum net power (kW)	X		X	3.6		If different from rated speed
3.2.3	Declared maximum torque speed (rpm)	X			2800		If applicable
3.2.3.1	Fuel delivery/stroke (mm ³) for diesel engine, fuel flow (g/h) for other engines, at maximum torque speed			X	1000		
3.2.3.2	Declared maximum torque (Nm)	X			12.0		If applicable

3.2.4	Declared 100% test speed	X			3600		If applicable
3.2.5	Declared Intermediate test speed	X			Not applicable		If applicable
3.2.6	Idle speed (rpm)	X			1600		If applicable
3.2.7	Maximum no load speed (rpm)	X			3800±50		If applicable
3.2.8	Declared minimum torque (Nm)	X			Not applicable		If applicable
3.3	Run-in procedure						Optional at choice of manufacturer
3.3.1	Run in time	X			3h		
3.3.2	Run in cycle	X			G2		
3.4	Engine test						
3.4.1	Specific fixture required: Yes/No	X			No		For NRSh only
3.4.1.1	Description, including photographs and/or drawings, of the system for mounting the engine on the test bench including the power transmission shaft for connection to the dynamometer:	X			Not applicable		
3.4.2	Exhaust mixing chamber permitted by manufacturer: Yes/No	X			No		For NRSh only
3.4.2.1	Exhaust mixing chamber description, photograph and/or drawing	X			Not applicable		If applicable
3.4.3	Manufacturers chosen NRSC: RMC/Discrete mode	X			Discrete mode		
3.4.4	Additional NRSC: E2/D2/C1	X			Not applicable		Only where additional cycles are declared in entries 1.12.1. or 1.12.2. of Part A
3.4.5	Number of pre-conditioning cycles prior to transient test	X			Not applicable		If applicable, minimum 1.0
3.4.6	Pre-conditioning for RMC NRSC: Steady-state operation/RMC:	X			Not applicable		If applicable, minimum 0.5
3.4.6.1.	In case of RMC, number of pre-conditioning RMC prior to RMC	X			Not applicable		Minimum 0.5

	NRSC test						
3.5	Lubrication system						
3.5.1	<i>Lubricant temperature</i>						If applicable
3.5.1.1	Minimum (deg. C)	X			-30		
3.5.1.2	Maximum (deg. C)	X			145		
3.6	Combustion Cylinder						
3.6.1	Bore(mm)			X	70		
3.6.2	Stroke(mm)			X	55		
3.6.3	Number of cylinders			X	1		
3.6.4	Engine total swept volume (cm ³)			X	212		
3.6.5	Swept volume per cylinder as % of parent engine			X	100%		If engine family
3.6.6	Volumetric compression ratio			X	8.66:1		Specify tolerance
3.6.7	Combustion system description			X	Spark ignition		
3.6.8	Drawings of combustion chamber and piston crown			X	Refer to drawing No.002 and 003		
3.6.9	Minimum cross sectional area of inlet and outlet ports (mm ²)			X	Inlet:393.5 Outlet:368.63		
3.6.10	<i>Valve timing</i>						
3.6.10.1	Maximum lift and angles of opening and closing in relation to dead centre or equivalent data			X	Maximum Lift of inlet: 5.70mm Maximum Lift of outlet: 5.75 mm; Angles of opening and closing Refer to drawing No.004		
3.6.10.2	Reference and/or setting range			X	In: 0.10-0.12mm Out: 0.15-0.17mm		
3.6.10.3	Variable valve timing system: Yes/No			X	No		If applicable and where intake and/or exhaust
3.6.10.3.1	Type: continuous/(on/off)			X	Not applicable		
3.6.10.3.2	Cam phase shift angle			X	Not applicable		
3.6.11	<i>Porting configuration</i>						2-stroke only, if applicable
3.6.11.1	position, size and number			X	Refer to drawing No.003		
3.7	Cooling system						Complete relevant section
3.7.1	<i>Liquid cooling</i>						

3.7.1.1	Nature of liquid			X	Not applicable		
3.7.1.2	Circulating pumps: Yes/No			X	No		
3.7.1.2.1	type(s)			X	Not applicable		
3.7.1.2.2	Drive ratio(s)			X	Not applicable		If applicable
3.7.1.3	Minimum coolant temperature at outlet (deg. C)	X			Not applicable		
3.7.1.4	Maximum coolant temperature at outlet (deg. C)	X			Not applicable		
3.7.2	<i>Air cooling</i>						
3.7.2.1	fan: Yes/No			X	No		
3.7.2.1.1	type(s)			X	Not applicable		
3.7.2.1.2	Drive ratio(s)			X	Not applicable		If applicable
3.7.2.2	Maximum temperature at reference point (deg. C)			X	248		
3.7.2.2.1	Reference point location			X	Spark plug washer		
3.8	Aspiration						
3.8.1	Maximum allowable intake depression at 100% engine speed and at 100% load (kPa)	X	X				
3.8.1.1	With clean air cleaner	X	X		2.0		
3.8.1.2	With dirty air cleaner	X	X		2.0		
3.8.1.3	Location, of measurement	X	X		Thermal baffle		
3.8.2	Pressure charger(s): Yes/No			X	No		
3.8.2.1	Type(s)			X	Not applicable		
3.8.2.2	Description and schematic diagram of the system (e.g. maximum charge pressure, waste gate, VGT, Twin turbo, etc.)			X	Not applicable		
3.8.3	Charge air cooler: Yes/No			X	No		
3.8.3.1	Type: air-air/air-water/other (specify)			X	Not applicable		
3.8.3.2	Maximum charge air cooler outlet temperature at 100% speed and 100% load (deg. C)	X	X		Not applicable		
3.8.3.3	Maximum allowable pressure drop across charge cooler at 100% engine speed and at 100% load (kPa)	X	X		Not applicable		
3.8.4	Intake throttle valve: Yes/No			X	No		
3.8.5	Device for recycling crankcase gases: Yes/No			X	No		

3.8.5.1	If yes, description and drawings				Not applicable		
3.8.5.2	If no, compliance with paragraph 6.10 of Annex VI to Commission Delegated Regulation (EU) (EU) 2022/992 on technical and general requirements: Yes/No	X			Not applicable		
3.8.6	<i>Inlet path</i>						2-stroke, NRS and NRSh only
3.8.6.1	Description of inlet path, (with drawings, photographs and/or part numbers)			X	Not applicable		
3.8.7.	<i>Air filter</i>			X			2-stroke, NRS and NRSh only
3.8.7.1	Type			X	170F/P		
3.8.8.	<i>Intake air-silencer</i>						2-stroke, NRS and NRSh only
3.8.8.1	Type			X	Not applicable		
3.9	Exhaust system						
3.9.1	Description of the exhaust system (with drawings, photos and/or part numbers as required)			X	Refer to drawing No.006		2-stroke, NRS and NRSh only
3.9.2	Maximum exhaust temperature (deg. C)	X			973		
3.9.3	Maximum permissible exhaust backpressure at 100% engine speed and at 100% load (kPa)	X	X		5.5		Note(2)
3.9.3.1	Location of measurement	X	X		The first chamber of muffler		
3.9.4	Exhaust backpressure at loading level specified by manufacturer for variable restriction after-treatment at start of test (kPa)	X			Not applicable		
3.9.4.1	Location and speed/load conditions	X			Not applicable		
3.9.5	Exhaust throttle valve: Yes/No			X	No		
3.10	Miscellaneous devices						
3.10.1	<i>Exhaust gas recirculation</i>				Not applicable		
3.10.1.1	Characteristics: cooled/uncooled, high pressure/low pressure/other (specify):			X	Not applicable		
3.10.2	<i>Water injection</i>				Not applicable		
3.10.2.1	Operation principle			X	Not applicable		

3.10.3.	Air injection				Not applicable		
3.10.3.1.	Operation principle:			X	Not applicable		
3.10.4.	Other(s)				Not applicable		
3.10.4.1.	Type(s):			X	Not applicable		
3.11	Exhaust after-treatment system						
3.11.1	<i>Location</i>		X		Not applicable		
3.11.1.1	Place(s) and maximum/minimum distance(s) from engine to first after-treatment device		X		Not applicable		
3.11.1.2	Maximum temperature drop from exhaust or turbine outlet to first after-treatment device (deg. C) if stated	X	X		Not applicable		
3.11.1.2.1	Test conditions for measurement	X	X		Not applicable		
3.11.1.3	Minimum temperature at inlet to first after-treatment device (deg. C), if stated	X	X		Not applicable		
3.11.1.3.1.	Test conditions for measurement:	X	X		Not applicable		
3.11.2	<i>Oxidation catalyst</i>				Not applicable		
3.11.2.1	Number of catalytic converters and elements			X	Not applicable		
3.11.2.2	Dimensions and volume of the catalytic converter(s)			X	Not applicable		
3.11.2.3	Total charge of precious metals			X	Not applicable		
3.11.2.4	Relative concentration of each compound			X	Not applicable		
3.11.2.5	Substrate (structure and material)			X	Not applicable		
3.11.2.6	Cell density			X	Not applicable		
3.11.2.7	Type of casing for the catalytic converter(s)			X	Not applicable		
3.11.3	<i>Catalytic exhaust gas after treatment system for NOx or three way catalyst</i>				Not applicable		
3.11.3.1	Type			X	Not applicable		
3.11.3.2	Number of catalytic converters and elements			X	Not applicable		
3.11.3.3	Type of catalytic action			X	Not applicable		
3.11.3.4	Dimensions and volume of the catalytic converter(s)			X	Not applicable		
3.11.3.5	Total charge of precious metals(mg)			X	Not applicable		
3.11.3.6	Relative concentration of each compound			X	Not applicable		

3.11.3.7	Substrate (structure and material)			X	Not applicable		
3.11.3.8	Cell density			X	Not applicable		
3.11.3.9	Type of casing for the catalytic converter(s)			X	Not applicable		
3.11.3.10	Method of regeneration	X		X	Not applicable		If applicable
3.11.3.10.1	Infrequent regeneration: Yes/No				No		If yes, complete section 3.11.6
3.11.3.11	Normal operating temperature range (deg. C)	X	X		Not applicable		
3.11.3.12	Consumable reagent: Yes/No			X	Not applicable		
3.11.3.12.1	Type and concentration of reagent needed for catalytic action			X	Not applicable		
3.11.3.12.2	Lowest concentration of the active ingredient present in the reagent that does not activate warning system (CD _{min}) (%vol)			X	Not applicable		
3.11.3.12.3	Normal operational temperature range of reagent		X		Not applicable		
3.11.3.12.4	International standard		X	X	Not applicable		If applicable
3.11.3.13	NOx sensor(s): Yes/No			X	No		
3.11.3.13.1	Type			X	Not applicable		
3.11.3.13.2	Location(s)			X	Not applicable		
3.11.3.14	Oxygen sensor(s): Yes/No			X	No		
3.11.3.14.1	Type			X	Not applicable		
3.11.3.14.2	Location(s)			X	Not applicable		
3.11.4.	<i>Particulate after-treatment system</i>				Not applicable		
3.11.4.1	Type of filtration: through flow/partial flow/wall flow/other (specify)			X	Not applicable		
3.11.4.2	Type			X	Not applicable		
3.11.4.3	Dimensions and capacity of the particulate trap			X	Not applicable		
3.11.4.4	Location place(s) and maximum and minimum distance(s) from engine		X		Not applicable		
3.11.4.5	Method or system of regeneration, description and/or drawing			X	Not applicable		
3.11.4.5.1	Infrequent regeneration: Yes/No			X	No		If yes, complete

							section 3.11.6.
3.11.4.5.2	Minimum exhaust gas temperature for initiating regeneration procedure (deg. C)			X	Not applicable		
3.11.4.6	Catalytic coating: Yes/No			X	No		
3.11.4.6.1.	Type of catalytic action:			X	Not applicable		
3.11.4.7	Fuel borne catalyst (FBC): Yes/No			X	No		
3.11.4.8	Normal operating temperature range (deg. C)			X	Not applicable		
3.11.4.9	Normal operating pressure range (kPa)			X	Not applicable		
3.11.4.10	Storage capacity soot/ash [g]			X	Not applicable		
3.11.4.11.	Oxygen sensor(s)				Not applicable		
3.11.4.11.1.	Type:				Not applicable		
3.11.4.11.2.	Location(s):				Not applicable		
3.11.5	<i>Other after-treatment devices</i>				Not applicable		
3.11.5.1	Description and operation			X	Not applicable		
3.11.6	<i>Infrequent Regeneration</i>				Not applicable		
3.11.6.1	Number of cycles with regeneration			X	Not applicable		
3.11.6.2	Number of cycles without regeneration			X	Not applicable		
3.11.7.	Other device(s) or feature(s)				Not applicable		
3.11.7.1.	Type(s):			X	Not applicable		
3.12	Fuel feed for liquid-fuelled CI or, where applicable, dual-fuel engines				Not applicable		
3.12.1	<i>Feed pump</i>				Not applicable		
3.12.1.1	Pressure (kPa) or characteristic diagram			X	Not applicable		
3.12.2	<i>Injection system</i>				Not applicable		
3.12.2.1	Pump				Not applicable		
3.12.2.1.1	Type(s)			X	Not applicable		
3.12.2.1.2	Rated pump speed (rpm)			X	Not applicable		
3.12.2.1.3	mm ³ per stroke or cycle at full injection at rated pump speed			X	Not applicable		Specify tolerance
3.12.2.1.4	Torque peak pump speed (rpm)			X	Not applicable		
3.12.2.1.5	mm ³ per stroke or cycle at full			X	Not applicable		Specify

	injection at torque peak pump speed						tolerance
3.12.2.1.6	Characteristic diagram			X	Not applicable		As alternative to entries 3.12.2.1.1 to 3.12.2.1.5
3.12.2.1.7	Method used: on engine/on pump bench			X	Not applicable		
3.12.2.2	<i>Injection timing</i>				Not applicable		
3.12.2.2.1	Injection timing curve			X	Not applicable		Specify tolerance, if applicable
3.12.2.2.2	Static Timing			X	Not applicable		Specify tolerance
3.12.2.3	<i>Injection piping</i>				Not applicable		
3.12.2.3.1	Length(s) (mm)			X	Not applicable		
3.12.2.3.2	Internal diameter (mm)			X	Not applicable		
3.12.2.4	Common rail: Yes/No			X	No		
3.12.2.4.1.	Type			X	Not applicable		
3.12.3	<i>Injector(s)</i>				Not applicable		
3.12.3.1	Type(s)			X	Not applicable		
3.12.3.2	Opening pressure (kPa)			X	Not applicable		Specify tolerance
3.12.4	<i>Electronic control unit (ECU): Yes/No</i>			X	No		
3.12.4.1	Type(s)			X	Not applicable		
3.12.4.2	Software calibration number(s)			X	Not applicable		
3.12.4.3	Communication standard(s) for access to data stream information: ISO 27145 with ISO 15765-4 (CANbased)/ ISO 27145 with ISO 13400 (TCP/IPbased)/ SAE J1939-73	X		X	Not applicable		
3.12.5	<i>Governor</i>				Not applicable		
3.12.5.1	Type(s)			X	Not applicable		
3.12.5.2	Speed at which cut-off starts under full load			X	Not applicable		Specify range, if applicable
3.12.5.3	Maximum no-load speed			X	Not applicable		Specify range, if applicable

3.12.5.4	Idle speed			X	Not applicable		Specify range, if applicable
3.12.6	Cold-start system: Yes/No			X	Not applicable		
3.12.6.1	Type(s)			X	Not applicable		
3.12.6.2	Description			X	Not applicable		
3.12.7	<i>Fuel temperature at the inlet to the fuel injection pump</i>				Not applicable		
3.12.7.1	Minimum (deg. C)	X			Not applicable		
3.12.7.2	Maximum (deg. C)	X			Not applicable		
3.13	Fuel feed for liquid fuel spark ignition engine						
3.13.1	<i>Carburetor</i>				mucan		
3.13.1.1	Type(s)			X	170F		
3.13.2	<i>Port fuel injection</i>						
3.13.2.1	single-point / multi-point			X	Not applicable		
3.13.2.2	Type(s)			X	Not applicable		
3.13.3	<i>Direct injection</i>						
3.13.3.1	Type(s)			X	Not applicable		
3.13.4	<i>Fuel temperature at location specified by manufacturer</i>						
3.13.4.1	Location	X			Not applicable		
3.13.4.2	Minimum (deg. C)	X			Not applicable		
3.13.4.3	Maximum (deg. C)	X			Not applicable		
3.14	Fuel feed for gaseous fuel engines or where applicable, dual fuel engines (in the case of systems laid out in a different manner, supply equivalent information)						
3.14.1	Fuel: LPG /NG-H/NG-L /NG-HL/LNG/Fuel specific LNG	X		X	Not applicable		
3.14.2	<i>Pressure regulator(s)/vaporiser(s)</i>				Not applicable		
3.14.2.1	Type(s)			X	Not applicable		
3.14.2.2	Number of pressure reduction stages			X	Not applicable		
3.14.2.3	Pressure in final stage minimum and maximum. (kPa)			X	Not applicable		
3.14.2.4	Number of main adjustment points			X	Not applicable		

3.14.2.5	Number of idle adjustment points			X	Not applicable		
3.14.3	Fuelling system: mixing unit/gas injection/liquid injection/direct injection			X	Not applicable		
3.14.3.1	<i>Mixture strength regulation</i>				Not applicable		
3.14.3.1.1.	System description and/or diagram and drawings			X	Not applicable		
3.14.4	<i>Mixing unit</i>				Not applicable		
3.14.4.1	Number			X	Not applicable		
3.14.4.2	Type(s)			X	Not applicable		
3.14.4.3	Location			X	Not applicable		
3.14.4.4	Adjustment possibilities			X	Not applicable		
3.14.5	<i>Inlet manifold injection</i>				Not applicable		
3.14.5.1	Injection: single-point/multi-point			X	Not applicable		
3.14.5.2	Injection: continuous/ simultaneously timed/ sequentially timed			X	Not applicable		
3.14.5.3	<i>Injection equipment</i>				Not applicable		
3.14.5.3.1	Type(s)			X	Not applicable		
3.14.5.3.2	Adjustment possibilities			X	Not applicable		
3.14.5.4	<i>Supply pump</i>				Not applicable		If applicable
3.14.5.4.1	Type(s)			X	Not applicable		
3.14.5.5	<i>Injector(s)</i>				Not applicable		
3.14.5.5.1	Type(s)			X	Not applicable		
3.14.6	<i>Direct injection</i>				Not applicable		
3.14.6.1	Injection pump/pressure regulator			X	Not applicable		
3.14.6.1.1	Type(s)			X	Not applicable		
3.14.6.1.2	Injection timing (specify)			X	Not applicable		
3.14.6.2	<i>Injector(s)</i>				Not applicable		
3.14.6.2.1	Type(s)			X	Not applicable		
3.14.6.2.2	Opening pressure or characteristic diagram			X	Not applicable		
3.14.7	<i>Electronic Control Unit (ECU)</i>				Not applicable		
3.14.7.1	Type(s)			X	Not applicable		
3.14.7.2	Adjustment possibilities			X	Not applicable		

3.14.7.3	Software calibration number(s)			X	Not applicable		
3.14.8	<i>Approvals of engines for several fuel compositions</i>				Not applicable		
3.14.8.1	Self-adaptive feature: Yes/No	X	X	X	Not applicable		
3.14.8.2	Calibration for a specific gas composition: NG-H/NGL/NG-HL/LNG/Fuel specific LNG	X	X	X	Not applicable		
3.14.8.3	Transformation for a specific gas composition: NGHT/NG-LT/NG-HLT	X	X	X	Not applicable		
3.14.9	<i>Fuel temperature pressure regulator final stage</i>				Not applicable		
3.14.9.1	Minimum (deg. C)	X			Not applicable		
3.14.9.2	Maximum (deg. C)	X			Not applicable		
3.15	Ignition system						
3.15.1	<i>Ignition coil(s)</i>				CQJY		
3.15.1.1	Type(s)			X	170F		
3.15.1.2	Number			X	1		
3.15.2	<i>Spark plug(s)</i>				DNNID		
3.15.2.1	Type(s)			X	F7TC		
3.15.2.2	Gap setting			X	0.7-0.8 mm		
3.15.3	<i>Magneto</i>			X			
3.15.3.1	Type(s)			X	Not applicable		
3.15.4	Ignition timing control: Yes/No			X	Yes		
3.15.4.1	Static advance with respect to top dead centre (crank angle degrees)			X	Refer to drawing No.005		
3.15.4.2	Advance curve or map			X	Refer to drawing No.005		
3.15.4.3	Electronic control: Yes/No			X	No		

Explanatory notes to Appendix 3:

(Footnote markers, footnotes and explanatory notes not to be stated on the information document)

- (1) As defined in Annex II to Commission Delegated Regulation (EU) 2022/992.
- (2) Refer to section 2.4.13. in Annex IX (engine family definition).

Declaration by manufacturer on compliance with Regulation (EU) 2016/1628

Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd. Hereby declares that the following engine type/~~engine family~~⁽¹⁾ complies in all respects with the requirements of Regulation (EU) 2016/1628 of the European Parliament and of the Council¹, Commission Delegated Regulation (EU) 2022/992 on technical and general requirements², Commission Delegated Regulation (EU) 2022/992 on monitoring of in-service engines³ and Commission Implementing Regulation (EU) 2022/992 on administrative requirements⁴ and does not use any defeat strategy. All emission control strategies comply, where applicable, with the requirements for Base Emission Control Strategy (BECS) and Auxiliary Emission Control Strategy (AECS) set-out in section 2 of Annex IV to Commission Delegated Regulation (EU) 2022/992 on technical and general requirements, and have been disclosed in accordance with that Annex and with Annex I of Commission Implementing Regulation (EU) 2022/992 on administrative requirements.

- 1.1 Make (trade name(s) of manufacturer) : SAIBOYUHU
- 1.2 Commercial name(s) (if applicable) : Not applicable
- 1.3 Company name and address of manufacturer : Chongqing Saibo RunTong Machinery Manufacturing Co., Ltd.
The second and third floors of No. 16, Yijian 2nd Village,
Qingmuguan Town, Shapingba District, Chongqing, China
- 1.4 Name and address of manufacturer's authorised representative (if any) : Liu Lu
MALGRATE (LC) VIA ROMA 62 CAP 23864
- 1.6 Engine type designation/~~engine family-~~
~~designation/FT~~⁽¹⁾ : ZT-170F/P

(Place): Chongqing, China

Date: 25th July. 2025

Signature:



Li Wei

(or visual representation of an 'advanced electronic signature' according to Regulation (EU) No 910/2014, including data for verification):

(Footnote markers, footnotes and explanatory notes not to be stated on the manufacturer's declaration)

(1) Strike out the unused options, or only show the used option(s).

Appendix-1

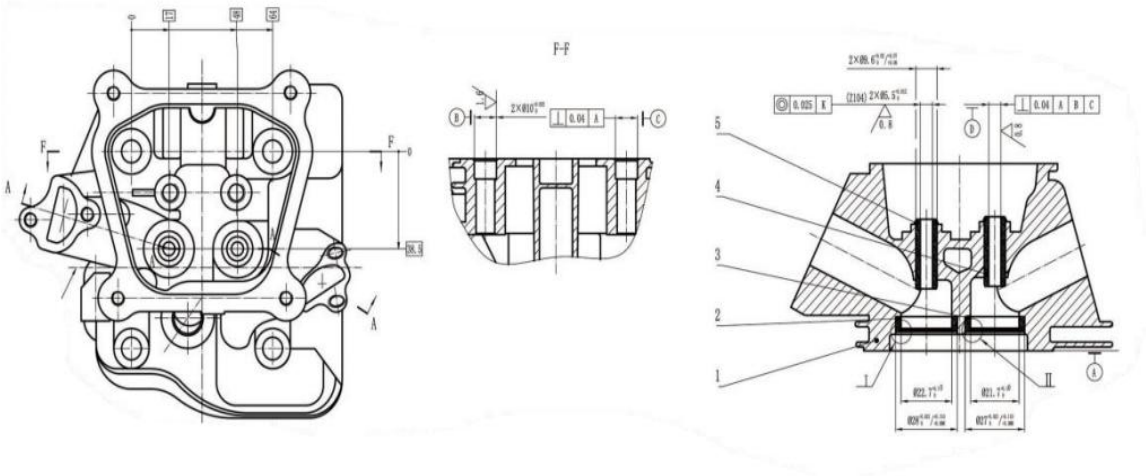
Drawings of engine



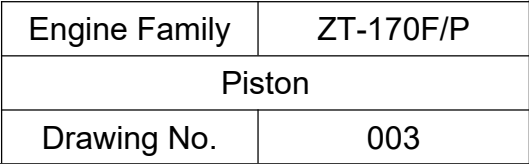
Trade name/trademark/manufacture name
Engine model
Engine serial number
Engine production date
Approval number/approval mark:
e49*2016/1628*XXXX/XXXXXXXXXX*XXXXX*00 or
e49*XXX/P V-XXX

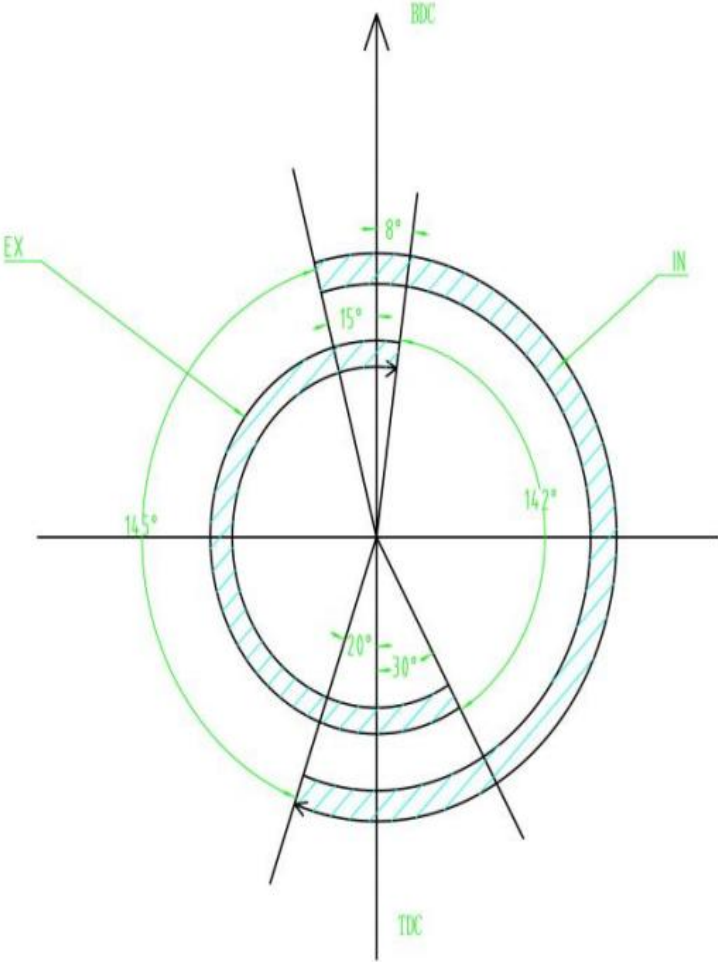
Remarks: this sample only shows the contents that need to be included on the engine marking,
the actual layout may adjust according manufacturer’s requirement.

Engine Family	ZT-170F/P
Position of statutory marking	
Drawing No.	001

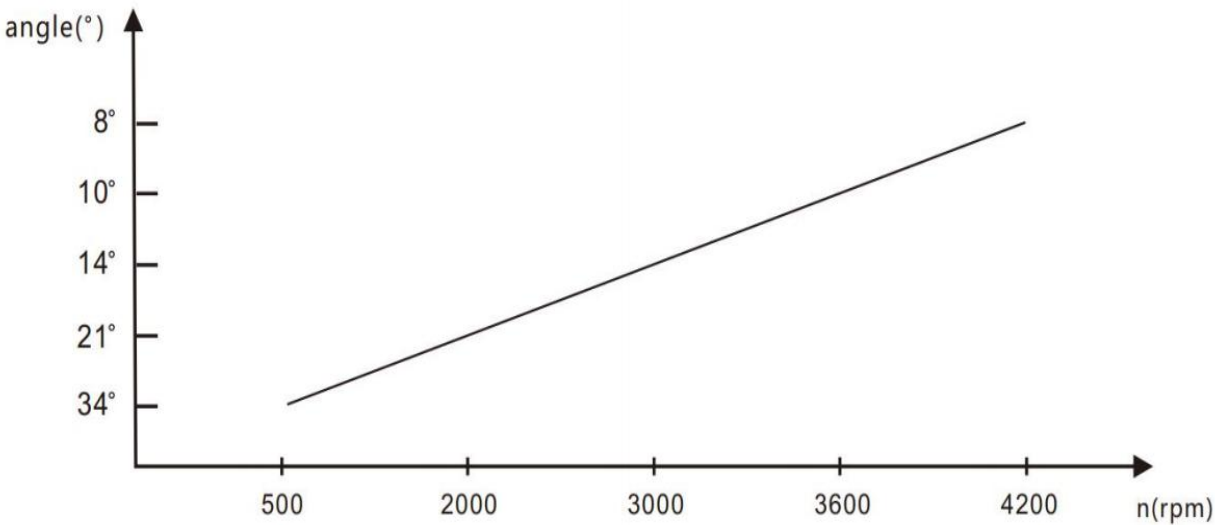


Engine Family	ZT-170F/P
Cylinder head	
Drawing No.	002

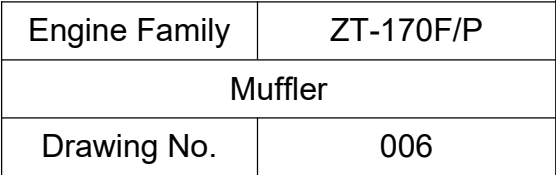


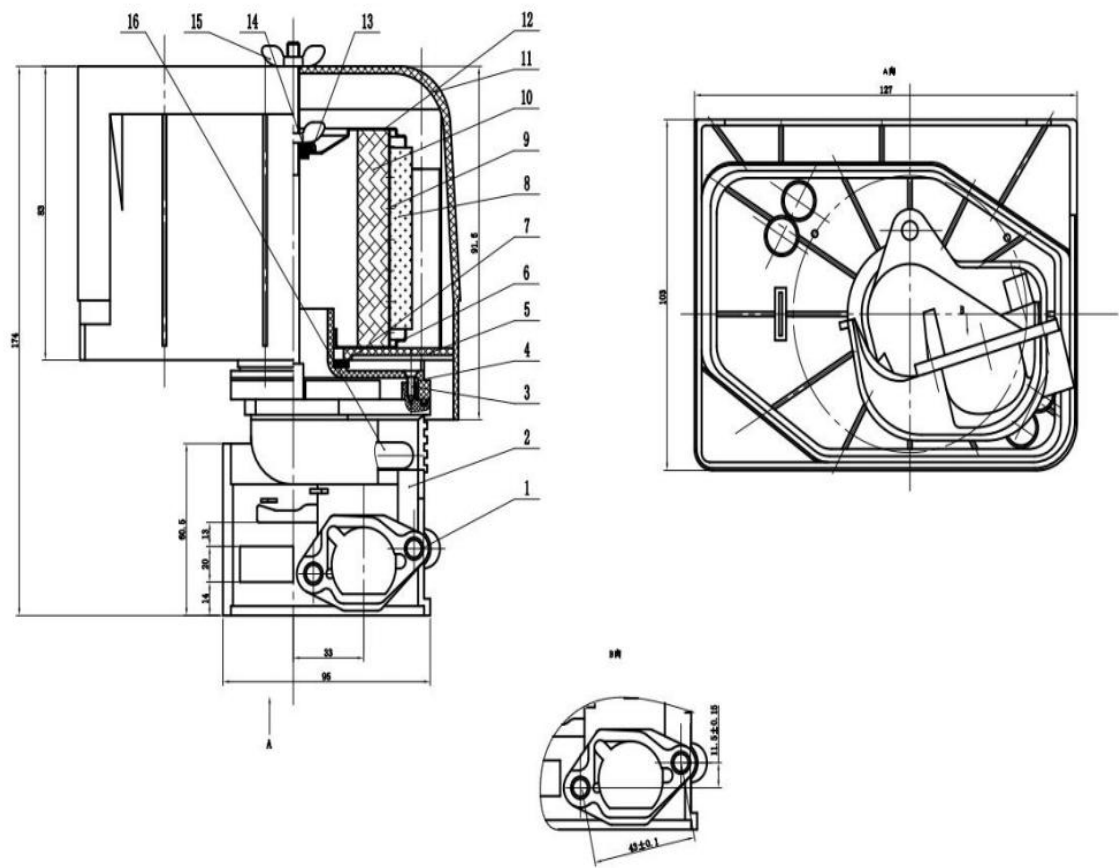


Engine Family	ZT-170F/P
Timing diagram	
Drawing No.	004



Engine Family	ZT-170F/P
Ignition advance curve	
Drawing No.	005





Engine Family	ZT-170F/P
Air filter	
Drawing No.	007